



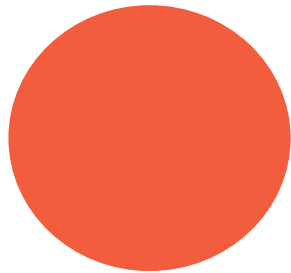
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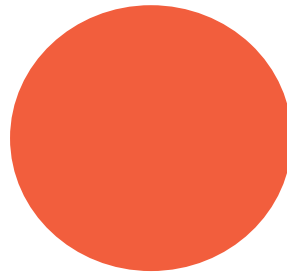
DEPARTMENT OF CSE
(ARTIFICIAL INTELLIGENCE & MACHINE LEARNING)

0th REVIEW

Drug Trafficking Detection on Social Media

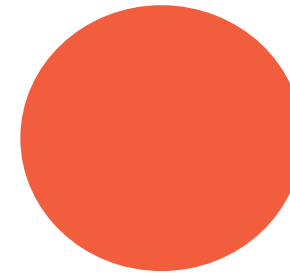


BATCH NO



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GUIDE



TABLE OF CONTENTS

- ***Introduction***
- ***Problem Identification***
- ***Problem Statement***
- ***Motivation***
- ***Objectives***
- ***Literature Survey***
- ***Base Paper***
- ***Proposed System***
- ***Dataset Description***
- ***Sample Dataset***
- ***Tools & Metrics***
- ***Merits & Demerits***
- ***Conclusion***



Introduction

Social media platforms have evolved into "direct-to-consumer" marketplaces for illicit substances. Traffickers exploit the anonymity of platforms like Instagram, Twitter, and Reddit to market controlled substances directly to users, often bypassing traditional security measures. This project proposes an automated AI tool to detect and hide these posts in real-time, acting as a digital shield for users against the online drug trade.



Problem Identification

*The core challenge in detecting online drug trafficking is the **"Needle in a Haystack"** problem, complicated by **"Algospeak"** (evasion tactics).*

Volume: *Identifying illicit posts is difficult due to the massive volume of benign daily chatter.*

Obfuscation: *Traffickers use emojis (e.g., 📄 for cocaine,) and slang to bypass traditional keyword filters.*

Ambiguity: *Simple filters fail to distinguish between a doctor discussing "painkillers" (Safe) and a dealer selling "painkillers" (Illicit).*



Problem Statement

Online drug dealing represents a critical public safety challenge characterized by high-volume digital communication and sophisticated linguistic obfuscation. Traditional manual oversight and static keyword lists are insufficient for detecting modern digital trafficking strategies. There is a lack of real-time, scalable tools that can automatically analyze social media text, decode drug slang, and prevent user exposure to illicit marketplaces.



Motivation

The motivation stems from the rising "Digital Opioid Crisis," where younger demographics are targeted by drug dealers on platforms they use daily. By automating detection, we can reduce the workload for law enforcement and prevent vulnerable users from accessing illegal distributors

Objectives

- **Develop an NLP Classifier:** Build a machine learning model (Naive Bayes/SVM) that accurately classifies text as "Illicit" or "Safe".
- **Emoji & Slang Decoding:** Implement a Semantic Mapping Layer that translates high-risk symbols (e.g.,  means "cocaine") into text features before analysis.
- **Real-Time Deployment:** Integrate the model into a Chrome Browser Extension that automatically hides/blurs suspicious posts as the user browses.



Literature Survey

Author / Year	Methodology	Key Insight
Li et al. (2019)	LSTM + ResNet	(Base Paper) Proved that text captions are the most reliable signal for detecting trafficking intent (95% F1-score). https://www.jmir.org/2019/6/e13803/
Alfatmi et al. (2025)	BERT + YOLO	Created "NarcoTrace," utilizing BERT for deep text analysis to categorize risk levels.
Sidorov et al. (2025)	LLMs (GPT)	Demonstrated that context-aware models outperform traditional keyword matching in understanding slang.



Base Paper

- **Base Paper**
 - **Title:** *Detection of Illicit Drug Dealers on Instagram (Li et al., 2019).*
 - **Methodology:**
 - *Analyzed 12,000+ posts using Deep Learning (LSTM).*
 - *Proved Text Captions detect dealers better than images.*
 - **Key Result:**
 - *Achieved 95% F1-Score in identifying sellers.*
 - **Limitation (The Gap):**
 - *Offline Analysis: It only analyzed old data, failing to stop users from seeing drugs.*
 - **Our Solution:**
 - *Real-Time Intervention: We use a Chrome Extension to block content instantly while browsing.*

Proposed System

The system follows a 3-step pipeline:

- **Input:** The Chrome Extension extracts text content from the user's active web page.
- **Preprocessing (The Core Innovation):**
 - **Slang Decoder:** A custom lexicon converts emojis and street terms into standard English (e.g., "needle  "cocaine").
 - **Vectorization:** TF-IDF converts the decoded text into numerical format.
- **Classification:** The ML Model (Naive Bayes) predicts the label. If "Drug-Related," the extension modifies the DOM to blur the post.

Dataset Description

- *We utilize a Hybrid Text Dataset to ensure robustness:*
- ***Trafficking Data (Class 1):*** Sourced from the Instagram Drug Dealer Detection repository (Mathison et al.) and augmented with synthetic data containing modern slang (e.g., "gas", "420").
- ***Safe Data (Class 0):*** Sourced from the UCI Drug Review Dataset, containing legitimate medical discussions to teach the model context (preventing false alarms on medical posts).



Sample Dataset

- Use this table to show the Problem (Ambiguity) and Slang to the reviewers. It proves your project is smart

Type	Raw Text Content	Decoded Feature (NLP)	Label
Trafficking	"Got that pure white  ready. DM for prices."	"Got that pure white [COCAINE] ready. DM for prices."	1 (Illicit)
Trafficking	"Blue M30s on deck.  Fast shipping."	"Blue M30s on deck. [SHIPPING] Fast shipping."	1 (Illicit)
Safe (Medical)	"My doctor prescribed Xanax for anxiety  ."	"My doctor prescribed Xanax for anxiety [PILL]."	0 (Safe)
Safe (Ambiguous)	"Look at the beautiful snow outside!   "	"Look at the beautiful snow outside! [COCAINE]"	0 (Safe)



Tools & Metrics

- **Tools & Metrics**
 - **Machine Learning & NLP (The Core):**
 - **Algorithm: Naive Bayes Classifier** (via Scikit-learn) for binary classification.
 - **NLP Tools: NLTK / SpaCy** for Slang Decoding, Tokenization, and Stop-word removal.
 - **Vectorization: TF-IDF** to convert text into numerical features.
 - **System Architecture:**
 - **Backend: Python (Flask)** – API that processes text in real-time.
 - **Frontend: JavaScript** (Chrome Extension) – Modifies the DOM to blur illicit posts.
 - **Evaluation Metrics:**
 - **Primary: F1-Score** – Crucial for handling imbalanced data (balances safety vs. false alarms).
 - **Secondary: Precision** (Avoids blocking medical info), **Recall** (Catches hidden dealers), **Confusion Matrix**.



Merits & Demerits

- **Merits (Pros):**
 - **Real-Time Protection:** Blocks content before the user reads it.
 - **Context-Aware:** Distinguishes between "selling drugs" and "medical advice."
 - **Scalable:** Monitors thousands of posts automatically.
- **Demerits (Cons):**
 - **Evolving Slang:** The slang dictionary must be updated frequently as dealers invent new codes.
 - **Encryption:** The tool is limited to public web pages and cannot scan private encrypted chats (e.g., WhatsApp).



Conclusion

- **Technological Bridge:** This project successfully overcomes the limitations of static keyword filters by integrating **NLP with Semantic Emoji Decoding**, effectively detecting the "Algospeak" used by modern traffickers.
- **Proactive Defense:** We shift the safety paradigm from "Reporting after harm" to "**Real-time Prevention**", utilizing a Chrome Extension to block illicit content before it reaches the user.
- **Data Strategy Success:** The use of a **Hybrid Dataset** (Real + Synthetic) proves effective in addressing the scarcity of illicit training data, ensuring the model is robust against both slang and medical ambiguity.
- **Societal Impact:** Ultimately, this tool acts as a scalable **Digital Shield**, protecting vulnerable demographics from exposure to drugs while significantly reducing the manual workload for content moderators.